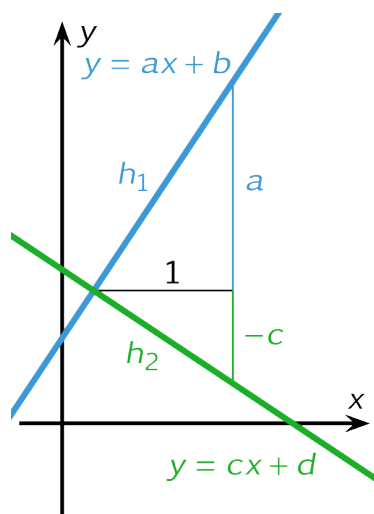


Formlen for hældninger mellem ortogonale linjer $a \cdot c = -1$

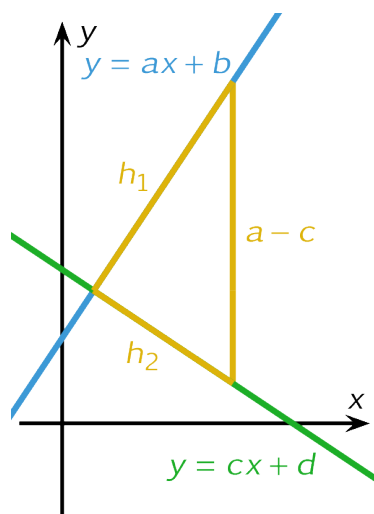
① Tegn figur:



② Beskriv de to retvinklede trekanter og opskriv udtryk for deres hypotenuuser:

$$h_1 = \sqrt{1^2 + (-c)^2} \quad \text{og} \quad h_2 = \sqrt{1^2 + a^2}$$

③ Tilføj den store trekant på figuren:



③ Opskriv udtryk for $(a - c)^2$ og ”isolér” ac :

$$(a - c)^2 = h_1^2 + h_2^2$$

$$(a - c)^2 = \left(\sqrt{1^2 + (-c)^2}\right)^2 + \left(\sqrt{1^2 + a^2}\right)^2$$

$$a^2 + c^2 - 2ac = 1^2 + (-c)^2 + 1^2 + a^2$$

$$a^2 + c^2 - 2ac = a^2 + c^2 + 2$$

$$-2ac = a^2 - a^2 + c^2 - c^2 + 2$$

$$ac = \frac{2}{-2}$$

$$ac = -1$$